

Zeus James Baltazar

work@zeusjames.com | ##### | #####
<https://www.linkedin.com/in/zeus0390> | <https://www.github.com/zeus0390>

Summary

Software Developer with hands-on experience in backend development, distributed systems, and automation. Proficient in systems programming, debugging, profiling, and performance optimization using primarily C++ and Python. Interested in building reliable, scalable, and maintainable systems, focusing on speed, performance and efficient design.

Professional Experience

Software Developer

Amdocs Philippines Inc.

August 2024 – June 2026

Taguig, Metro Manila, Philippines | Hybrid

- Investigated the core codebase and leveraged modern C++ skills to resolve bugs, submitting fixes to the core development team.
- Led the end-to-end design and development of workflow automation scripts using Python, Groovy, Bash, and KornShell in Linux environments, eliminating manual intervention and increasing operational efficiency and developer productivity.
- Optimized SQL scripts for efficient data retrieval from Oracle and PostgreSQL, then tested and deployed them to production.
- Collaborated with cross-functional Scrum teams to implement software changes and features, using Git, Perforce, Jira, and Confluence to coordinate development work.

Project

Peer-to-Peer File Sharing System

<https://www.github.com/zeus0390/p2p-file-sharing-system>

- Developing a peer-to-peer file-sharing system using C++ for distributed communication on Linux.
- Built the core networking layer as a reusable library with dual-stack IPv4/IPv6 support using TCP socket programming.
- Implemented concurrent connection and operation handling using an event-driven I/O multiplexer with epoll.
- Applied modern C++ practices including the RAII, smart pointers, move semantics, STL, synchronization, and multithreading.
- Designed and implemented a custom application-layer protocol for reliable data transmission over a byte stream.
- Developed file upload and download over peer-to-peer TCP connections.

Adaptive Time-Window Chunking with Threshold-Bound Querying

- Developed a Python-based automation tool to efficiently retrieve large datasets ranging from hundreds of thousands to millions of records from the company's Elasticsearch API while adhering to the 10,000-record query limit per request.
- Peer-reviewed by senior developers, tested, and deployed to production for ongoing operational and development use.
- Implemented a binary-search-style algorithm with dynamic sliding time windows to automatically calculate near-maximum threshold-compliant date/time ranges for data extraction, eliminating manual adjustments and achieving ~99.9% utilization of the API query limit while maintaining complete data retrieval.

Technical Skills

Programming & Scripting Languages: C, C++, Python, Bash, SQL

Programming Concepts: Object-Oriented Programming (OOP), Memory Management, Templates, Concurrency, Multithreading

Debugging & Profiling: GDB, GProf, AddressSanitizer (ASan), ThreadSanitizer (TSan)

Build & Version Control: CMake, Git

Networking: TCP/IP, Socket Programming, SSH

Operating Systems: Unix, Linux, Windows

Development Tools: Vim, Docker

Education

Bachelor of Science in Computer Engineering

Technological Institute of the Philippines - Quezon City

May 2018 – September 2023

GWA: 1.72/5.00 (1.00 = highest)